Final Project Implementation Roadmap

Topic 1: A Low-complexity Data-driven Learning to Navigate Satellites in Low Earth Orbit (LEO)

Instructor: Dr. Sirani M. Perera

Final Submission Date: December 7, 2026

Team Size: 2 Members

Required Report Length: 7 Pages (excluding cover & references)

PROJECT ALIGNMENT & SCOPE STRATEGY

This 4-week execution roadmap aligns Topic 1 with the core syllabus of Datta's *Numerical Linear Algebra and Applications*. Instead of black-box deep learning, the project uses data-driven matrix methods—specifically least-squares estimation, QR/SVD factorization, condition number bounds, and iterative solvers—to construct a fast, robust LEO satellite navigation algorithm.

TEAM WORKLOAD DISTRIBUTION

Member 1: Aerospace Engineering Student

- **Orbital Mechanics:** Formulates linearized state-space matrices for LEO orbital dynamics.
- **Data Modeling:** Generates realistic noisy GPS and altitude sensor observation vectors.
- **Error Analysis:** Assesses state estimation accuracy against orbital propagation constraints.
- **Documentation:** Authors Introduction, Orbital Physics derivations, and Methodology.

Member 2: Computer Science Student

- **Algorithm Design:** Implements QR, SVD, and iterative least-squares solvers in MATLAB/Python.
- **Complexity Optimization:** Analyzes FLOP counts, sparsity patterns, and computational speedup.
- **Numerical Stability:** Performs perturbation analysis and evaluates condition numbers.
- **Documentation:** Formats numerical graphs, code appendices, and performance benchmarks.

COURSE CURRICULUM MATRIX MAPPING

The project directly leverages specific course modules: **Sections 4.3–4.6** (Conditioning & Perturbation Analysis), **Sections 7.1–7.8** (Householder QR Factorization & SVD), and **Sections 8.1–8.5** (Least-Squares Solutions & Pseudoinverse).

4-WEEK TACTICAL TIMELINE

Week 1: Problem Formulation & Orbital Data Modeling

Phase 1

- **Aerospace:** Derive simplified two-body linearized orbital equations of motion for LEO satellites ($r \approx 400\text{ km}$). Define state vector $x(t) = [r, v]^T$.
- **Computer Science:** Set up Python/MATLAB execution environment and git repository. Formulate system matrix structure $Ax \approx b$.
- **Joint:** Construct synthetic trajectory dataset with simulated Gaussian sensor noise.

Milestone Deliverable: Mathematical formulation document and synthetic data generator script.

Week 2: Numerical Solver & Matrix Factorizations

Phase 2

- **Computer Science:** Implement baseline least-squares solver via Normal Equations, QR Factorization (Householder), and SVD (Datta Ch. 7 & 8).
- **Aerospace:** Validate calculated position/velocity outputs against ground-truth orbital trajectories.
- **Joint:** Compare numerical accuracy between standard matrix inversion and orthogonal factorizations.

Milestone Deliverable: Executable solver script with initial accuracy comparison figures.

Week 3: Complexity Reduction & Error Analysis

Phase 3

- **Computer Science:** Derive FLOP count for solvers; exploit matrix structure/sparsity to minimize execution time.
- **Aerospace:** Perform perturbation analysis (Datta Ch. 4). Test model sensitivity against observation noise spikes.
- **Joint:** Benchmark execution speed vs. accuracy for low-complexity satellite deployment scenarios.

Milestone Deliverable: Benchmarking tables, flop calculations, and error-bound plots.

Week 4: Report Compilation & Rubric Alignment

Phase 4

- **Aerospace:** Write Introduction, Historical Background, and Methodology derivations.
- **Computer Science:** Generate final high-resolution plots, format MATLAB/Python code appendix, write Results.
- **Joint:** Format 7-page document (Times New Roman 11pt, 1.0 spacing, 1-inch margins) and verify rubric compliance.

Milestone Deliverable: Final 7-Page Project Report ready for Canvas upload.

RUBRIC CHECKLIST COMPLIANCE

Core Requirements Verified: Exact 7-page limit maintained • Full theoretical derivation of solvers • Quantitative numerical results & graphs • Complete code included in Appendix (does not count toward page limit) • Clear objective-result connection.